Linear Equations & Systems

Section 1: Multi-Step Linear Equations & Distribution

1 If $4(2x - 3) + 7 = 3(x + 5) + 2$, what is the value of x?

a. 3

b. 4

c. 5

d. 6

2 If $\frac{3x - 5}{4} = \frac{x + 7}{2}$, what is the value of x?

a. 19

b. 9

c. 11

d. 15

3 If $3x - 2y = 14$ and $x = 2y - 2$, what is the value of xy?

a. 12

b. 18

c. 24

d. 30

Section 2: Systems of Linear Equations

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4 In the system of equations $5x + 3y = 23$ and $2x - 3y = 5$, what is the value of $x + y$?

a. 4

b. 5

c. 6

d. 7

5 A system of linear equations is given by $3x + 4y = 18$ and $ax + 8y = 36$. For what value of a does the system have infinitely many solutions?

a. 3

b. 6

c. 8

d. 12

6 For what value of k does the system $kx - 6y = 10$ and $2x - 3y = 7$ have no solution?

a. 2

b. 4

c. -4

d. 6

Section 3: Coefficients, Constants & Equivalent Forms

7 If $ax + b = 5x + 12$ is true for all values of x, what is the value of ab?

a. 17

b. 30

c. 60

d. 120

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8 If $3(2x + a) = bx + 15$ has infinitely many solutions, what is the value of $a + b$?

a. 9

b. 11

c. 15

d. 21

9 In the equation $\frac{1}{2}(4x - 6) + k = 2x + 5$, for what value of k does the equation hold for all real numbers x?

a. 2

b. 5

c. 8

d. 11

Section 4: Real-World Modeling & Rate Scenarios

10 A local theater sells adult tickets for 12 dollars each and student tickets for 8 dollars each. If a total of 150 tickets were sold for 1,560 dollars, how many student tickets were sold?

a. 60

b. 70

c. 80

d. 90

11 A plumber charges a fixed diagnostic fee of 65 dollars plus 45 dollars per hour of repair work. If a customer received a total bill of 290 dollars, how many hours did the repair take?

a. 4

b. 5

c. 6

d. 7

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12 Two printing presses, Press A and Press B, print copies at constant rates. Press A prints 40 copies per minute and Press B prints 60 copies per minute. If Press B begins 10 minutes after Press A, after how many minutes of Press B running will both presses have printed the same total number of copies?

a. 15

b. 20

c. 25

d. 30

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